



STRATHMORE INSTITUTE OF MATHEMATICAL SCIENCES
MASTER OF SCIENCE IN DATA SCIENCE & ANALYTICS
ASSIGNMENT
DSA 8301: Statistical Inference for Big Data

DUE DATE: 17th June 2026

Assignment Instructions

1. Ensure that you submit your assignment by the due date. Late submissions will NOT be accepted.
2. Your assignment should be well-organized, clearly written, and free of grammatical errors. Use appropriate headings and subheadings to structure your report.
3. All code should be well-commented to explain the purpose of each section and the logic behind your analysis.
4. Include all relevant outputs, such as tables, figures, and statistical summaries,

Assignment Objective

The purpose of this assignment is to enable students to apply statistical inference techniques learned in this course to a real-world dataset using Python. Students are expected to perform exploratory data analysis, parametric statistical methods, nonparametric statistical methods, and interpret the results in the context of the chosen problem.

Part A: Dataset Selection

Select a publicly available dataset and conduct a comprehensive statistical analysis using Python. The analysis should demonstrate your understanding of both parametric and nonparametric statistical methods covered in the course.

Select a dataset satisfying the following requirements:

- At least 100 observations.
- At least four variables.

- Contains both continuous and categorical variables.
- Obtained from a reputable source.

Possible sources include:

- Kaggle
- UCI Machine Learning Repository
- World Bank Open Data
- WHO Data Repository
- Government Open Data Portals

Examples of suitable datasets include:

- Student performance data.
- Healthcare datasets.
- Housing price datasets.
- Agricultural production datasets.
- Financial market datasets.
- Environmental and climate datasets.

Part B: Data Description and Exploratory Data Analysis

Perform a detailed exploratory analysis of the selected dataset.

Required Tasks

1. Describe the dataset:
 - Source of the data.
 - Number of observations.
 - Number of variables.
 - Description of each variable.
2. Data preprocessing:
 - Identification and treatment of missing values.
 - Detection of outliers.
 - Data consistency checks.
3. Compute descriptive statistics:
 - Mean.
 - Median.
 - Variance.
 - Standard deviation.

- Range.
- Interquartile range.

4. Produce graphical summaries:

- Histograms.
- Boxplots.
- Scatterplots.
- Correlation heatmaps.

5. Assess distributional assumptions:

- Histograms and density plots.
- Q-Q plots.
- Shapiro-Wilk normality test.

Note: All results must be interpreted appropriately.

Part C: Parametric Statistical Analysis

Apply at least **three** parametric statistical methods.

Possible methods include:

1. Confidence intervals.
2. One-sample t-test.
3. Two-sample t-test.
4. Paired t-test.
5. Analysis of Variance (ANOVA).
6. Linear regression analysis.
7. Maximum likelihood estimation.

For each method:

- Clearly state the research question.
- Formulate null and alternative hypotheses.
- Verify assumptions.
- Present test statistics.
- Report p-values.
- Interpret the findings.
- Draw appropriate conclusions.

Part D: Nonparametric Statistical Analysis

Apply at least **three** nonparametric statistical methods.

Possible methods include:

1. Sign test.
2. Wilcoxon signed-rank test.
3. Mann-Whitney U test.
4. Kruskal-Wallis test.
5. Spearman rank correlation.
6. Kolmogorov-Smirnov test.
7. Kernel density estimation.
8. Bootstrap confidence intervals.

For each method:

- Clearly state the research question.
- Formulate null and alternative hypotheses.
- Explain why a nonparametric approach is appropriate.
- Present test statistics.
- Report p-values.
- Interpret the findings.
- Draw appropriate conclusions.

Python Requirements

Students are expected to make appropriate use of Python packages, including but not limited to:

- `pandas`
- `numpy`
- `matplotlib`
- `seaborn`
- `scipy`
- `statsmodels`
- `scikit-learn`

Submission Requirements

Each student shall submit the following:

1. A technical report in PDF format.
2. Python script (.py) or Jupyter Notebook (.ipynb).
3. The dataset used for the analysis.

Report Structure

The report should contain the following sections:

1. **Title Page** (Include the title of the study, student's name, registration number, course name, lecturer's name, institution, and date of submission.)
2. **Introduction** (Provide background information on the problem under investigation, explain the motivation for selecting the dataset, state the objectives of the analysis, and outline the statistical questions to be addressed.)
3. **Dataset Description** (Describe the source of the dataset, the number of observations and variables, definitions of all variables used in the analysis, data types, and any data cleaning or preprocessing procedures undertaken.)
4. **Methodology** (Describe the statistical methods used in the analysis. Explain the exploratory data analysis procedures, parametric methods, nonparametric methods, assumptions of the techniques employed, and the Python packages used to implement the analysis.)
5. **Results** (Present the findings obtained from the analysis. Include tables, figures, descriptive statistics, hypothesis test results, confidence intervals, model outputs, and other relevant statistical summaries. Results should be clearly organized and appropriately labelled.)
6. **Discussion** (Interpret the results in the context of the study objectives. Explain the meaning of the findings, compare outcomes from parametric and nonparametric methods, discuss any unexpected results, and comment on the practical implications of the analysis.)
7. **Conclusions** (Summarize the major findings of the study, state the conclusions drawn from the statistical analyses, discuss limitations of the work, and provide recommendations for future studies or further analysis.)
8. **References** (List all sources consulted in conducting the study, including datasets, books, journal articles, websites, and software documentation. References should follow a consistent citation style such as APA, Harvard, or IEEE.)

Recommended report length: **5–10 pages**.

End of Assignment